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Paper

Neck Posture Monitoring System Based on Image Detection and Smartphone Sensors Using the Prolonged Usage Classification Concept

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Since the beginning of the smartphone era, the number of smartphone users has increased dramatically. Consequently, several aspects of their use have provoked concerns from researchers. From a health-related perspective, text neck syndrome is a consequence of excessive smartphone use. Therefore, we introduce a neck posture monitoring system to help prevent text neck syndrome in smartphone users. The system operates on smartphones using the embedded rotation vector sensors and the camera for image detection. Data from the sensors and camera are used to calculate the neck angle. The results of an experiment show that the calculation algorithm is not significantly different from the photogrammetric method, which was proposed earlier as a method to measure the neck angle from a side view image. In summary, we propose a system that includes the detection and classification of the user's neck posture to raise awareness and help promote better posture to prevent text neck syndrome among smartphone users. © 2018 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

Keywords: text neck; smartphone usage; posture monitoring; image detection; classification

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1. Introduction

According to a report on smartphone ownership by the Pew Research Center [1], from 2013 to 2015, the median smartphone ownership in emerging and developing nations climbed from 21 to 37%, while that in advanced economic nations is steady at 68%. Meanwhile, South Korea holds the top rank for smartphone ownership, as 88% of the population owns a smartphone and the rest 12% own at least a cell phone. The top ten nations for smartphone ownership include other advanced countries such as Australia, Israel, America, Spain, and the United Kingdom.

From this growing usage of smartphones, multiple concerns have arisen from the healthcare perspective. Previously, as the usage of computer increased, syndromes such as computer vision syndrome (CVS) became prevalent in the population. This syndrome is also a concern with regard to excessive smartphone use. Other syndromes, such as cellphone elbow, blackberry thumb, and text neck syndrome have been mentioned in addition to CVS.

One medical study [2] has suggested that, overall, a person may spend approximately 1400 h per year bending his or her neck forward in the reading position and this could be up to 4000 h per year for teenagers. A person who remains in this position for a prolonged period could develop severe neck pain. We discuss the details of this study in Section 2.1; nonetheless, this study indicates that this reading position is similar to the position for smartphone usage.

In this paper, we propose a system based on the smartphone platform to monitor the neck posture of users while using a smartphone. The system is based on the smartphone platform to enable system mobility. It consists of two major parts: first, the calculation of the neck angle, and second, the classification of the neck posture based on the neck angle and the usage time. In the neck angle calculation part, we use the rotation vector sensor embedded in the smartphone to calculate the tilt angle, while the image detection technique is used to calculate the face angle between the smartphone plane and user's face plane. The face angle allows the system to overcome limitations in calculating the neck angle with higher accuracy when the user's face plane is not parallel to the smartphone plane. Then, the system uses these two attributes, namely the tilt angle and the face angle, to calculate the neck angle based on the proposed neck angle calculation formula. Later in this paper, we evaluate the proposed algorithm and formula by conducting an experiment on 30 human subjects to compare the neck angle from our proposed smartphone-based system with the previously published photogrammetric method [3]. The evaluation was done using a statistical analysis to calculate the mean differences in the results from these two methods. Furthermore, we discuss a solution for classifying the neck posture based on the user's neck angle and usage time. This, linked to the prolonged usage concept, helps to increase users' awareness not only that they might be in a bad position but also that they might be using their smartphone for too long.

2. Related Studies

In this section, we survey studies that are related to our work. We first discuss studies that concern health-related issues due to the use of smartphones and how to handle them. Then we examine studies that relate technically to our methodology, that is, activity recognition and image detection.

2.1. Health-related issues of smartphone The occurrence of CVS [4] has been emphasized by several researches since

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the introduction of the personal computer. This syndrome is caused by a low blink rate, which is likely to occur when users are looking at any type of monitor for an extended period. A low blink rate will eventually lead to dry eyes and cause CVS. Symptoms of CVS may include blurred vision, headache, neck pain, fatigue, and eye strain. One study showed that of a total of 795 students from five universities in Malaysia, 89.9% had suffered at least one symptom of CVS [5]. This study also showed that, of the people who suffered from CVS, most spent more than 2 h per day on a computer.

Multiple studies have raised the importance of talking the issue by using technologies to prevent users from having CVS [6–8]. Each study took a different approach in countering the issue but, in general, these works demonstrate a good example in preventing health issues with cutting-edge technology.

One of the latest emerging health-related issues for smartphones is text neck. Text neck is a syndrome that causes severe neck pain and eventually leads to severe headaches [2]. Treatment for this serious condition may include physical therapy. Therefore, a preventive approach to this syndrome is recommended. This study has also suggested that the weight felt by the cervical spine increases exponentially with increasing neck angle. This study measured the weight with neck angles of 0°, 15°, 30°, 45°, 60°, and 90°. The cervical spine experiences approximately 4.5 kg of weight when the neck angle is in the neutral position. This weight can increase to up to 27 kg when the neck angle is 60°, and the weight cannot be measured when the neck is bent completely forward to 90°.

Two studies have proposed solutions to measure the neck angle using wearable sensors [9,10]. These studies included the implementation of a sensor in the earphone and a stationed image processing device. Nonetheless, these proposals did not provide accurate measurements of the neck movement or angle. Smart Pose [11] is one of the latest studies attempting to find a preventive solution for this syndrome. The authors of Smart Pose developed a system based on the sensors embedded in modern smartphones. Smart Pose collects data and assumes that the user's neck angle is the same as the smartphone angle, which is determined by the embedded smartphone accelerometer and gyroscope. Even though this is a very robust approach, there exists the possibility that a user's face plane is not exactly parallel to the smartphone plane. Therefore, this limitation is still present in their system.

2.2. Activity recognition In this section, we discuss activity recognition technology. There are many applications and systems that have been developed to monitor users' activities and to help promote healthier behavior. Some of these systems are based on chronic syndromes, while others are related to the quality of life. An overview of activity recognition and its development offers several perspectives on this subject [12,13]. Activity recognition can be performed using sensors embedded in modern devices such as accelerometers, compasses, gyroscopes, and barometers. Another study has provided an interesting review of a sensor network capable of monitoring activity [14]. That study outlined the possibility of implementing multiple sensors for health monitoring as well as the importance of the architecture of the system. In terms of real-time monitoring, another study has proposed real-time activity monitoring using a combination of a wristband and a smartphone [15]. This study did not focus on any purpose or activity; instead, it pointed out that such a system would help increase self-awareness and eventually improve the user's well-being. While all the previously mentioned studies focused on activity monitoring, Raffaele proposed a framework to handle data from all the sensors in the cloud [16]. This framework supports a 24/7 service of activity monitoring for both online and offline systems, providing better interactions between the cloud and sensors in the body area network as well as the management of the overall system. In general, most studies pointed out the possibility of using sensors embedded in portable devices to monitor activities and raise awareness of one's health. This led us to the use of both the embedded rotation vector sensors and camera in the smartphone, which will be discussed in detail in Section 3.

2.3. Image detection For image detection, we focus here on studies that are related to face detection with a mobile device similar to a smartphone. Several studies have suggested the use of a Haar-like feature classifier because such classifiers have been proven to be accurate and fast [17]. Haar is a feature-based cascade classifier originally proposed by Viola and Jones [18]. An implementation of Haar-like feature classifiers has been demonstrated in a system to detect the drowsiness and distraction level of a driver using a smartphone camera [19], a system to detect fatigue [20], and a smartphone photo software program for face detection [21]. In addition to these studies, a Haar-like feature classifier was used in another system for advance face feature recognition [22]; however, this system did not process the image locally but used a cloud service. Therefore, additional computational resources were needed for the system. Conversely, another famous algorithm for image detection is the local binary pattern (LBP) [23]. The working mechanism of LBP is similar to that of Haar; however, LBP uses integers in its calculations while Haar uses floating point numbers. As a result, LBP consumes much less resources and allows shorter computational times. Nonetheless, there is a tradeoff in the accuracy for LBP.

In this section, we discussed the syndromes mentioned in Section 1 and demonstrated how related studies had used available technology to overcome or prevent these syndromes. This shows that the consequences of using modern technology cannot be ignored. Focusing on developing an efficient prevention system for the text neck syndrome, we analyzed published studies to explore the possibility of implementing a neck posture monitoring system on the smartphone platform using embedded sensors and image detection techniques. Our system, which considers both the neck angle and the usage time when classifying the neck posture, can help users to understand and prevent unhealthy neck postures in the future.

3. Methodology

3.1. System overview Figure 1 shows an overview of the system. The system has been designed according to machine-to-machine (M2M) architecture [24]. The main components of the architecture are M2M devices, an M2M gateway, and an M2M cloud service.

In our case, the smartphone works as the M2M device and M2M gateway. The sensors and camera embedded in the smartphone are responsible for collecting data from the user, while the network communication capability of the smartphone allows it to communicate with various services on the cloud via the Internet.

The classifier training process for face feature detection had been done earlier, prior to real-time system operation. After the monitoring process starts, the system initiates three processes at the same time: face feature detection, phone tilt angle detection, and usage time logging. Face feature detection uses the previously trained classifier. After it detects the facial features, which are the face, eyes, and mouth, it calculates the face angle based on the detection. The neck angle calculation uses the face angle and the phone tilt angle as input attributes for the process. Then, the system classifies the neck posture using the calculated neck angle and usage time. The system sends the result of the classification back to the users as feedback on their smartphone usage behavior. This information is also stored for weekly and monthly summary to give the users a better idea of their overall behavior.

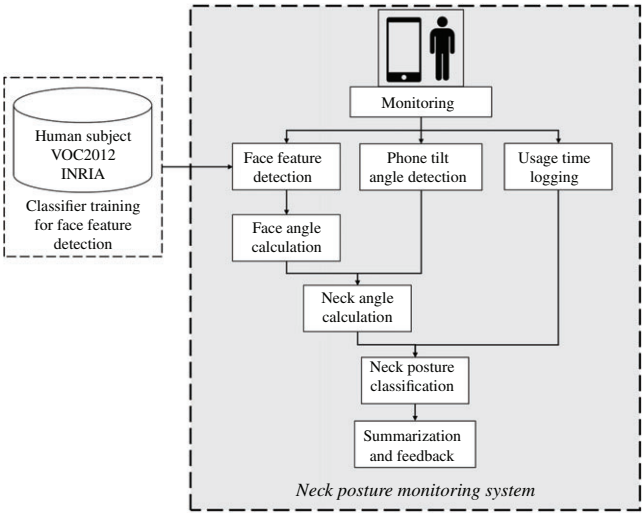
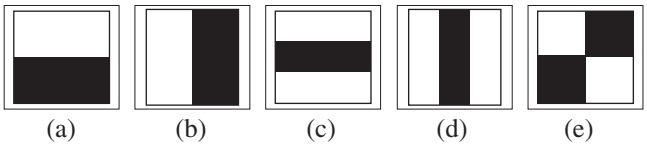


Fig. 1. Overview of the proposed neck posture monitoring system

3.2. Face feature detection Face feature detection allows the system to detect the face, eyes, and mouth. This detection process is crucial to our algorithm because it enables the calculation of the face angle. The process makes use of the OpenCV library, which provides the system with multiple detection algorithms. The two best known algorithms for detecting facial features are the Haar-like feature algorithm and the LBP algorithm.

Both Haar-like feature and LBP use features in detecting sum of pixels for object of interest. Figure 2 shows the five features that were used in the training process for the sum-of-difference calculation within each feature. This was done by determining the difference between the pixel density in the black area and the that in the white area. In practice, the window is defined to move along an image. In each window, each feature at all possible sizes is applied for the sum-of-difference calculation. For example, a window size of 24×24 pixels could result in more than 160 000 features in total (43 200, 43 200, 27 600, 27 600, and 20 736 features for each type of features in Fig. 2(a)–(e), respectively). As this process can result in high computational time, AdaBoost and cascade approach were applied to reduce the overall computational time. The main difference between Haar and LBP is that Haar uses floating point numbers in the calculation while LBP uses integers. In our case, during the training process, the sum of differences within the object area (face, eyes, or mouth) was calculated for all positive images. Then, during the detection process, the classifier performed the calculation on each feature within one window size for detection. Thus, if the result of the calculations matched with what the classifier learned from the training process, then it would classify the object within that window as a positive object. All in all, neither Haar nor LBP was specifically looking for the corner of the objects, which in our case were the face, eyes, and mouth; instead, they focused on calculating the overall pixel density within the defined window using the five features. Note that the window size also changed during this process to handle different object sizes in each image. As a result, the detected object size might vary between detections. In order to tackle this problem, we proposed the use of an averaged calculation, which will be discussed further in Section 3.5. Figure 3 shows an example how each feature can be used to calculate the sum of difference of pixel density when assuming that the red rectangular box is the window. Note that in one window each feature will change to all possible sizes for the calculation.

The training of the classifier for both the Haar-like feature algorithm and the LBP algorithm was conducted on a total sample



(a),(b) : Edge features
(c),(d) : Linear features
(e), : Four-rectangle features

Fig. 2. Five Haar-like features used in image detection

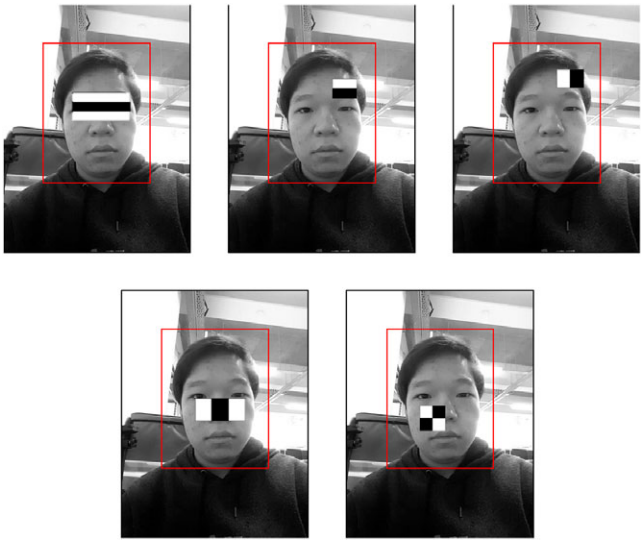


Fig. 3. Example of using five Haar-like types of features within a window

of 3742 images. We obtained this training set from human subjects, the INRIA database [25], and VOC2012 [26]. The training set consisted of 900 positive images and 2842 negative images, where the positive images were images with the objects of interest, which in our case were the face, eyes, and mouth, and the negative images were images without the objects of interest. The result of the training gave us a model for face, eyes, and mouth detection, which was essential for the face angle calculation.

In order to avoid false detection of face features during the process, we first reduce our region of interest (ROI) for face feature detection. The system attempts to detect only the face from the image. If the system is unable to detect the face, it assumes that the user is not currently using or looking at the smartphone. If the system successfully detects the face, it will narrow down the ROI for the detection of other face features by cropping only the face area of the image. Thus, it is easier for the mouth and eyes classification model to avoid any false positive detection. In our previous study [27], we performed an experiment to compare the performance of the two algorithms Haar and LBP with 500 images, where the objects of interest (human face with eyes and mouth) were present in all 500 images. Table I shows the results, which include the detection accuracy and its false negative detection value. From a total of 500 images used in that experiment, the false positive within the ROI area (face area) is less then 3% in all cases. Note that there is no result for 'False Positive within ROI' for face detection, as face detection was used to define the ROI of each image. In terms of detecting the correct positions of the outer canthi and left and right corners of the mouth, the system used the averaging approach, which we discuss further in Section 3.5. The results also showed that a combination of the two algorithms works best in terms of accuracy and calculation speed. In addition, it was suggested that the system should implement the Haar-like

Table I. Comparison of Haar and LBP classification model performance

Classifier	Algorithm type	Detection time (s)	True positive	False negative	False positive	False positive within ROI
Face	Haar	578.02	487	13	79	—
Face	LBP	155.99	429	71	77	—
Mouth	Haar	284.11	490	10	25	3
Mouth	LBP	143.20	486	14	75	13
Eye	Haar	211.78	457	43	57	5
Eye	LBP	145.67	343	157	45	10

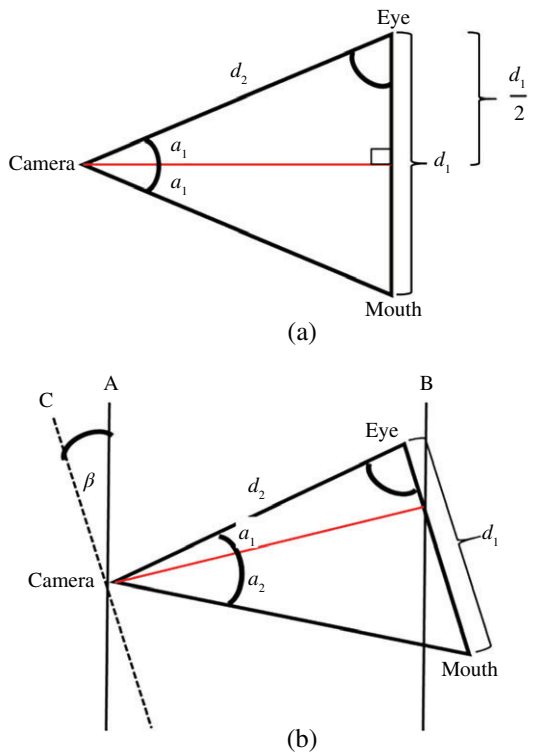


Fig. 4. Example scenarios for calculating the face angle. (a) The scenario where face pane is parallel to the smartphone pane. (b) The scenario where face pane is pitching toward the smartphone pane

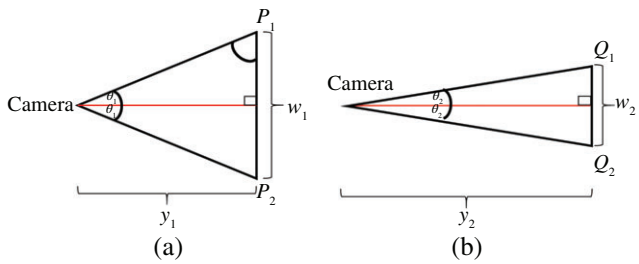


Fig. 5. Changes in object width at different distances between the object and the camera. (a) Example of object width at closer distance to the camera. (b) Example of object width at farther distance to the camera

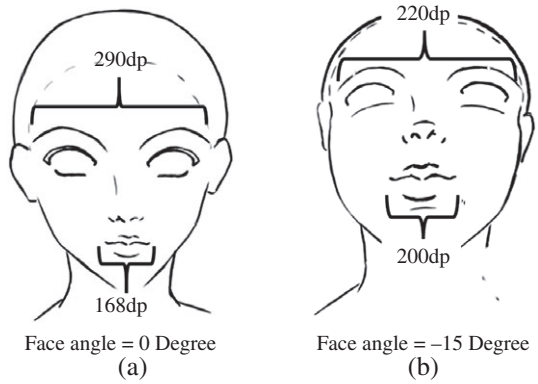


Fig. 6. Example of different face ratios at different face angles. (a) Face angle $= 0^\circ$. (b) Face angle $= -15^\circ$

feature classifier algorithm for face and eyes detection because it provided significantly higher accuracy than LBP. Conversely, the system should implement the LBP algorithm for mouth detection because it provided similar accuracy to the Haar algorithm but performed better in terms of calculation speed.

3.3. Face angle To calculate the user's neck angle, the system first obtains a face angle value. The face angle is the angle between the cervical spine and the imaginary line of the smartphone screen. This is done using trigonometry. Figure 4 shows two example scenarios for calculating the face angle. In Fig. 4(a), the camera plane, which refers to the imaginary line of the smartphone screen, is exactly parallel to the face plane, where the distance from the eyes to the camera is equal to d_2. On the other hand, Fig. 4(b) shows a situation where the face is pitching slightly forward toward the camera while the camera remains in the same position and the user's eye gaze is perpendicular to the camera plane. Lines A and B are parallel, and line C is parallel to the line drawn from the eyes to the mouth. In this case, the distance d_2 between the eyes and the camera is shorter than that in the previous scenario, and the distance between the camera and the mouth is longer. The angle β in the Fig. 4(b) represents the face angle used in the calculation for the neck angle. In this scenario, line A represents the smartphone screen, and the angle β can be calculated using trigonometry.

In our calculation, both the distances between the eyes and mouth to the smartphone cannot be determined. Therefore, we need to use the face ratio approach. If the user's face is pitching forward toward the screen, the distance between the eyes and the smartphone screen decreases, while the distance between the mouth and the screen increases. Figure 5 shows the differences in the widths of two objects from the top view when the distance between the object and the camera changes. As an example, in Fig. 5(a), the object could be a user's face from the top view. Let w_1 be the distance (width) from P_1 to P_2, w_2 be the distance (width) from Q_1 to Q_2, and y_1 and y_2 be the distances from the camera to the corresponding objects; the visual angle, which is the angle $\theta_1 + \theta_1$ in Fig. 5(a), is larger than the angle $\theta_2 + \theta_2$ in Fig. 5(b), and the value w_1 is considered to be larger than w_2 [28]. As a result, we can calculate the face angle using the face ratio. Figure 6 shows an example of face ratio differences when looking from different camera angles. Figure 6(a) shows the width of the eyes as 290 dot pixels at a face angle equal to 0°. Meanwhile, Fig. 6(b) shows that the width of the eyes is reduced to 220 dot pixels when the face angle is -15°.

Table II. Example of calculated neck angle value from one subject over 10 consecutive seconds

Second no.	Average (°)	Minimum (°)	Maximum (°)	Standard deviation	Variance
1	35.71	32.13	39.40	1.91	3.66
2	34.75	30.30	39.40	2.03	4.12
3	36.21	34.73	38.67	1.15	1.32
4	29.49	25.45	33.98	2.24	5.02
5	20.94	19.10	25.11	2.20	4.84
6	23.42	17.25	26.51	2.41	5.80
7	21.06	18.13	26.46	2.04	4.16
8	21.59	18.10	26.82	1.97	3.88
9	21.13	18.38	27.47	2.19	4.80
10	22.36	18.62	26.69	2.13	4.54

Equation (1) shows the calculation of the face angle, where *DFR* represents the default face ratio. *DFR* is the default face ratio of each user when the face plane is parallel to the camera plane. This value is obtained when the user trains the application. This process occurs when the user first uses the application. The system asks the user to train the application by taking five pictures with the user's line of sight perpendicular to the screen. The system calculates the face ratio of each image by dividing the width of the upper part of the face (the distance between eyes) by the width of the lower part of the face (the width of the mouth). Then it calculates the average of the five images and stores this value in the database for later use. The current face ratio (*CFR*) is the face ratio obtained from the current picture taken from the smartphone's front camera. The value 0.05 is the percentage value, that is, every five percentages of face ratio change equals one degree of face angle change. The value 0.05 was obtained from an empirical study based on the data from a total of 100 images of four face angle values, -15°, 0°, 15°, and 30°. If the face is pitching backward from the screen, the *CFR* will be less than the *DFR*. Therefore, the face angle will have a negative value. Conversely, if the face is pitching forward toward the smartphone screen, the *CFR* will be greater than the *DFR*. Therefore, the face angle will have a positive value. The unit of the face angle is in degrees. One of the benefits of using the face ratio when calculating the face angle is that it is scale-invariant. The distance between the smartphone screen and the user's face does not affect the ratio because the widths of both the upper and lower parts of the face change in the same direction. Therefore, the ratio remains the same.

$$FaceAngle = \frac{CFR - DFR}{0.05} \tag{1}$$

3.4. Smartphone tilt angle One of the attributes we use to calculate the neck angle is the smartphone tilt angle. The system obtains the smartphone tilt angle value using a virtual sensor called the rotation vector sensor. The rotation vector sensor is a software-based sensor that consists of an accelerometer and a magnetic field sensor [29]. This sensor has been incorporated by Google into all Android-enabled devices. The angle can be obtained for all axes (X, Y, and Z); however, in our case, we are interested in the angle between the Y- and Z-axes. We can retrieve this value using the provided Android API.

3.5. Neck angle calculation Our system calculates the neck angle using the two attributes that we mentioned earlier: the face angle and the smartphone tilt angle. A preliminary study of this calculation with only one sample human subject in 2015 [30] showed promising results. However, during that study, the quality of the device used to obtain the images and the coding technique needed several improvements. Therefore, in this study, we performed the calculation based on the same formula proposed in our previous study but with improved results due to the optimization of the face feature detection attributes and the neck angle calculation attributes. Equation (2) shows the formula for calculating the neck angle, where *PhoneAngle* is the phone tilt angle value obtained from the rotation vector sensor, *FaceAngle* is the calculated face angle value mentioned in Section 3.3, and the variable n represents the number of photos used in the calculation. In our case, OpenCV does not provide a constant result in the detection. Thus, the system calculates the neck angle on 30 sets of data (photos and sensor values) and averages them. By doing so, it reduces the errors that might occur from false detection and other noise. The number 30, which is our n value, comes from the frame rate setting of the video obtained from the smartphone camera. Therefore, each averaged neck angle value represents 1 s of video. In general, the system calculates the neck angle 30 times in 1 s, but it will only record the final average value and use the value as the neck angle for that particular instant. This approach is taken in order to reduce the variation as much as possible even if there are no or slight changes of the extracted face features between each frame. Thus, the averaged value of the neck angle, which is used to represent each second of the video, is more reliable than the neck angle calculated from a single frame of each second. Table II shows an example of neck angle calculation and its variation from one subject for 10 consecutive seconds in degrees. The table demonstrates the average, minimum, maximum, standard deviation, and variance of 30 neck angle values that were calculated in 1 s.

$$NeckAngle = \frac{\sum_{i=1}^{n}(PhoneAngle + FaceAngle)}{n} \tag{2}$$

3.6. Scenarios Figure 7 demonstrates all the possible scenarios of the calculation. Scenario 1 (Fig. 7(a)) shows the situation where the face plane is perfectly parallel to the smartphone plane. In this case, we can conclude that the user's neck angle is equal to the smartphone tilt angle, which is the value obtained from the smartphone sensor. However, this situation rarely occurs in daily usage.

Figure 7(b) demonstrates the situation where the face plane is pitching backward away from the smartphone plane. In this case, the neck angle will be less than the smartphone tilt angle. Therefore, to correctly calculate the neck angle, we use the face angle value, as mentioned in Section 3.3. The *FaceAngle* for this situation will have a negative value. Therefore, after calculating the neck angle from (2), the *NeckAngle* value will be less than the *PhoneAngle* value.

In the third scenario, as shown in Fig. 7(c), the face plane is pitching forward toward the smartphone plane. Opposite to the scenario in Fig. 7(b), the calculated *FaceAngle* will have a positive value. As a result, the calculation using (2) will give a *NeckAngle* value that is greater than the *PhoneAngle*. Figure 8 shows a sample

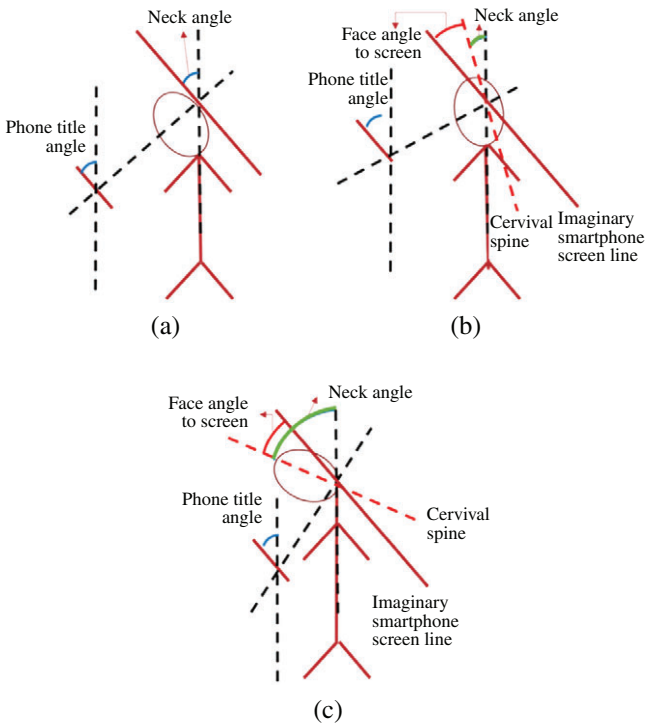


Fig. 7. Three neck angle calculation scenarios. (a) Face pane is parallel to the smartphone screen. (b) Face pane is pitching backward away from the smartphone screen. (c) Face pane is pitching forward toward the smartphone screen

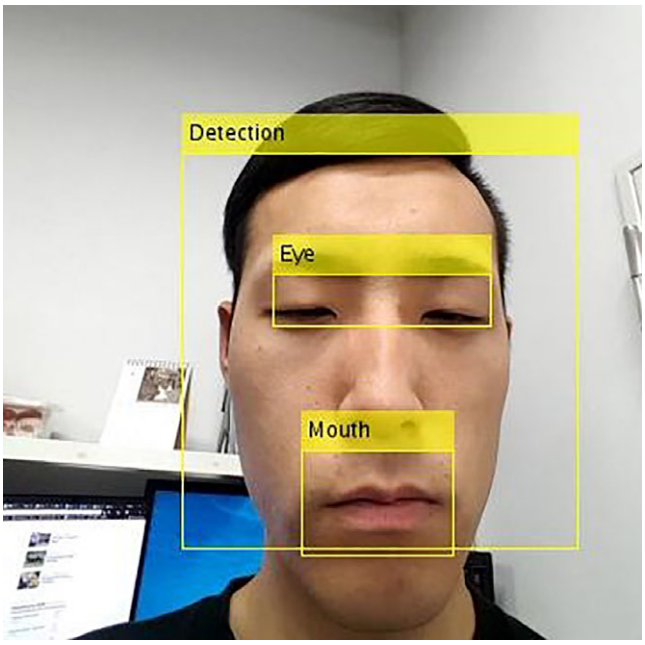


Fig. 8. Example of image detection for the face, eyes, and mouth

image from the experiment where the system detects the face, eyes, and mouth and uses them to calculate the face angle and neck angle.

3.7. Neck angle classification For this classification, we propose a classification method based on the neck angle in our previous study. This solution was solely based on the angle and did not consider other factors such as the usage time. The classification based on the angle is the same, where the angle ranges were 0–15°, 15–30°, 30–45°, 45–60°, and more than 60°. The classes are healthy, normal, slightly unhealthy, unhealthy, and very unhealthy for each respective angle range.

However, there is evidence that one of the causes of neck pain is related to prolonged smartphone usage [31]. Therefore, to make the classification more effective, the amount of usage time should also be considered. According to the recommendation for avoiding office syndrome [32], making a movement every 30 min could prevent severe pain in the back and neck. We make use of this recommendation in our classification by dividing the 30-min length into five periods, each period lasting 7.5 min. Longer usage times is unhealthy in the classification. The usage times are classified such that 0–7.5 min is healthy, 7.5–15 min is normal, 15–22.5 min is slightly unhealthy, 22.5–30 min is unhealthy, and more than 30 min is very unhealthy. The main purpose of this division was to provide ease of visualizing feedback to the user, and the five periods of time division was to match the five classes while still presenting the fact that the less the user uses the smartphone, the less chances they would have for suffering from text neck syndrome.

We combined the two concepts of the classification, and Table III shows a classification diagram with both the angle and the usage time. The angle-based concept first classifies the neck posture; then, the result worsens based on the usage time, which is logged by the system at the start of the monitoring process. If the usage time is considered healthy, the result will remain the same, while if the usage duration is considered very unhealthy, the result will worsen by four steps to the very unhealthy class in all cases.

In addition, we consider that the user is using the smartphone when the system successfully detects face features and calculates the neck angle. Thus, if the neck angle is continuously detected, the system will constantly record the usage time and perform the classifications based on the pre-defined conditions given in Table III. However, the system will stop recording when it cannot detect face features for neck angle calculation. In other words, the system assumes that when this happens, the user has looked away or making changes in his/her position. Thus, it will stop recording the usage time, and the prolong usage duration will be reset to zero. In general, the limitation of this concept is that it cannot handle the position changes while the face features are still being detected, as this would require further devices for activity recognition.

4. Experiment and Demonstration

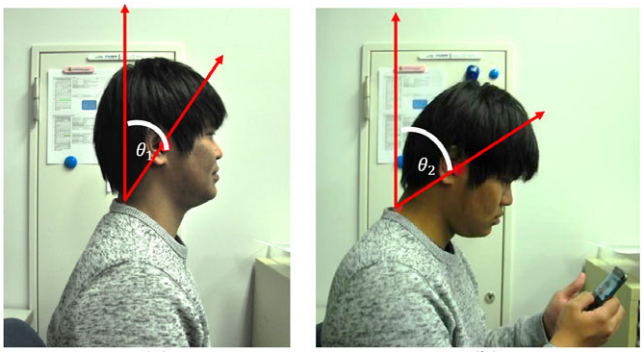
In this section we discuss an experiment we conducted to validate the proposed algorithm. This section includes the design of the experiment, the evaluation method, the statistical analysis, and the experimental results.

4.1. Experiment design We collected data from 30 human subjects, which consisted of 20 males and 10 females of six different nationalities. The age range of the subjects was between 20 and 31 years. We informed all subjects of the experimental procedures, and all subjects signed a consent form. The experiment was conducted in an office setting where the average brightness was 320 lx. This brightness level was also the recommended lighting condition for normal office and classroom [33].

The experiment process started after the subjects had signed the consent form. First, we collected a frontal face image with a camera; we used this picture to determine each individual subject's face ratio. Then, we placed two red sticker markers on the subject, one on the right tragus and the other on the large spinous process of the C7 vertebra. We used these markers to determine the neck angle via the photogrammetric method, which we will discuss later in this paper. The tragus was easily visible, and the spinous process of the C7 vertebra was located by running a hand along the back until reaching a large bump. Then, we asked the subject to sit with

Table III. Prolonged usage classification diagram

	Duration (min)				
Neck anglers (°)	0–7.5 healthy	7.5–15 normal	15–22.5 slightly unhealthy	22.5–30 unhealthy	>30 very unhealthy
0–15 healthy	Healthy	Normal	Slightly unhealthy	Unhealthy	Very unhealthy
15–30 normal	Normal	Slightly unhealthy	Unhealthy	Very unhealthy	Very unhealthy
30–45 slightly unhealthy	Slightly unhealthy	Unhealthy	Very unhealthy	Very unhealthy	Very unhealthy
45–60 unhealthy	Unhealthy	Very unhealthy	Very unhealthy	Very unhealthy	Very unhealthy
>60 very unhealthy	Very unhealthy	Very unhealthy	Very unhealthy	Very unhealthy	Very unhealthy



(a) (b)

Fig. 9. Example of neck angle calculation by using the photogrammetric method. (a) Subject's sitting position for reference angle. (b) Subject's sitting position duing experiment

their back straight and look straight ahead without bending their neck forward. We used a camera to take a picture to measure the reference angle for the photogrammetric method, which we discuss in Section 4.2.

The subject started the experiment by watching a 1-min-long video on a smartphone in a neutral viewing position according to the subject's preference. During the video playback, the smartphone and the video camera recorded the video simultaneously. These two video sources were later used for the neck angle calculation and evaluation, respectively. Lastly, we performed a statistical analysis on the obtained data.

4.2. Photogrammetric method This method makes use of the neck angle measured from a side view photo of a person. This method has been proven to be reliable [3]; therefore, in this paper, we used this method to validate the neck angle calculated from our smartphone monitoring system.

We recorded a video using a video camera, which was set up perpendicular to the subject. The video resolution was 640×480 pixels. The frame rate of the video was 24 frames per second. However, to synchronize the video data with the video file from the smartphone, we considered the first frame of each second to be representative of the 1-s window. We used this representative frame as data for the photogrammetric method and calculated the neck angle based on that data. Figure 9 shows an example of the images used to calculate the neck angle using the photogrammetric method. Figure 9(a) shows an image of a subject sitting straight up; the angle θ_1 is used as the reference angle for this subject. Figure 9(b) shows an image of the subject while undergoing the experiment. The angle θ_2 is the angle measured by the photogrammetric method. To calculate the neck angle, we calculated the difference between the two angles, i.e. $\theta_2 - \theta_1$. Figure 10 shows the positions of two red stickers that were used as a reference point for photogrammetric method. The result of this calculation was used later in the statistical analysis comparing these results with those obtained via the smartphone method.

Fig. 10. Positions of two red stickers for photogrammetric method for neck angle measurement

4.3. Statistical analysis Using a total of 30 subjects, we collected 1770 frames of data for both the smartphone application and the photogrammetric method. To verify that the results were reliable, we performed a paired sample t-test on the calculated neck angle from the smartphone system and the photogrammetric method to prove that there was no significant difference between the two methods. Equations (3) and (4) show the null hypothesis and the alternative hypothesis of this test, where μ_1 and μ_2 in both equations represent the mean of the neck angles from the smartphone monitoring system and the photogrammetric method, respectively. The null hypothesis represents that the difference of the two mean values $(\mu_1 - \mu_2)$ is equal to zero, and the alternative hypothesis represents the opposite. If the null hypothesis is not rejected, it means that the average difference of the neck angle from smartphone monitoring system and photogrammetric method is not large enough to be statistically significant.

$$H_0 : \mu_1 - \mu_2 = 0 \tag{3}$$

$$H_1 : \mu_1 - \mu_2 \neq 0 \tag{4}$$

To perform a paired sample t-test on these two sets of data, we calculated the difference between each frame of data. Figure 11 shows a histogram of the difference between the two methods in each frame of data. The data were assumed to be normally distributed, which was one of the prerequisites for performing the paired sample t-test. Therefore, we performed the Kolmogorov–Smirnov test on the data with a significance level (α) equal to 0.05. The p-value from the test was 0.798. Because the p-value was greater than the significance value, we concluded that the data were normally distributed. Figure 12 shows a graph of the test, where the blue line represents the empirical cumulative distribution function and the red line represents standard normal cumulative distribution function.

Table IV shows the descriptive statistics of the two sets of data. The photogrammetric method calculated a mean neck angle

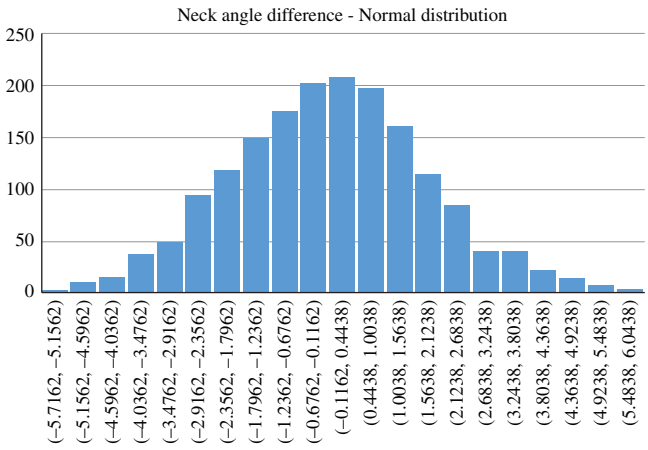


Fig. 11. Normal distribution of the difference value from the two neck angle measurements

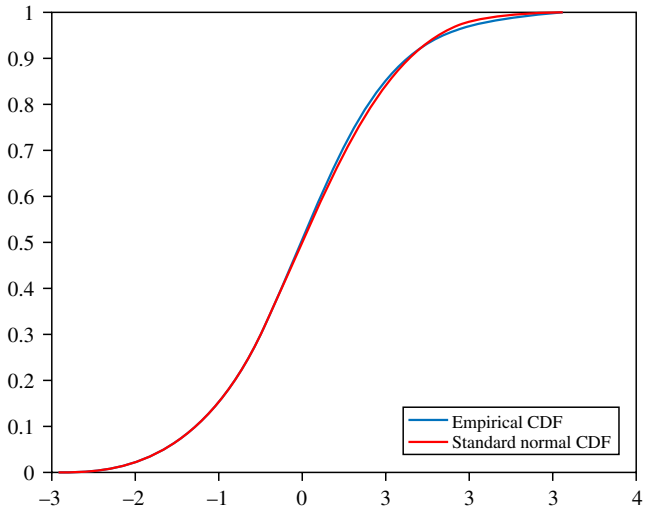


Fig. 12. Kolmogorov–Smirnov test plot

Table IV. Descriptive statistical analysis of the data from the two measurement methods

Statistical attributes	Photogrammetric method	Smartphone monitoring system
Mean	16.72	15.78
Median	15.94	16.29
Standard deviation	7.25	7.29
Variance	52.65	53.15
Data count	1770	1770

from the 30 subjects of 16.72°, while the smartphone application calculated a mean of 15.78°. The standard deviations are 7.25° and 7.29° for photogrammetric method and the smartphone application, respectively.

Table V shows the result of a paired sample t-test on the two sets of data. Equation (5) shows the formula for calculating the t-value for the paired sample t-test where $\overline{d}$ is the mean difference of the two sets of data and $SE(\overline{d})$ is the standard error of the mean difference. The test was performed with a confidence level of 95% ($\alpha = 0.05$). The degree of freedom for the test was 1769. Therefore, the t-critical values for the two-tailed analysis were − 1.961 and 1.961. The t-value for the analysis was −1.373 and the p-value

Table V. t-Test for two-paired sample mean results

Statistical attributes	Value
Alpha	0.05
Population mean of difference ($\overline{d}$)	0.063
Standard error of mean difference SE($\overline{d}$)	0.046
95% confidence interval	−0.154, 0.027
Degree of freedom	1769
Pearson correlation	0.964
t-value	−1.373
p-value	0.170
t-Critical for two-tailed analysis	−1.961, 1.961
Observation count	1770

was equal to 0.170, which was greater than the confidence level we set at 95% ($\alpha = 0.05$).

$$t = \frac{\overline{d}}{SE(\overline{d})} \tag{5}$$

4.4. Experiment results From the results, the t-value for the paired sample t-test is −1.373, which is between the t-critical values for the two-tailed analysis (−1.961 and 1.961). This leads us to accept the null hypothesis and conclude that the neck angle measured by the smartphone monitoring system has produced a result that is as accurate as the photogrammetric method at a confidence level of 95%. In other words, the paired sample t-test showed that the neck angle calculated would lie within the 95% confidence interval of mean difference, which was between −0.154° and 0.027° of differences. Note that during the experiment where we collected the data from the subjects with smartphone, the subjects were not required to look at the smartphone perpendicularly; instead, they were able to use the smartphone freely in their comfort position. As a result, the experiment showed that even if the subject did not maintain perpendicular eyesight to the smartphone, the system was capable of accurately calculating the neck angle, where it showed no significant difference when compared to the result from photogrammetric method.

4.5. Demonstration After the neck angle calculation was evaluated in the experiment, we performed as demonstration of the classification concept with a volunteer human subject. The subject was asked to use a smartphone for 30 min. The subject could move the phone freely to his comfort position as well as look away from the smartphone screen or take a short break within this 30-min period of the experiment. As mentioned in Section 3.7, the system tracked the usage time based on the detection of face features. Figure 13 shows the usage time of the subject over 30 min. The usage time increased as long as the system was capable of detecting face features and calculating neck angles. Then, the system stopped tracking the usage time when it failed to detect the face features. As an example, the usage time was constantly increasing until the 15th minute mark, when it failed to detect the face features and with an assumption that the subject had stopped using the smartphone. As a result, the usage time was reset to zero and the system restarted the usage time tracking process when it successfully detected the face features again.

Figure 14 shows the calculated neck angle plotted together with the classification result where the primary vertical axis (left vertical axis) shows the neck angle in degree and the secondary vertical axis (right vertical axis) shows the class label (1 = Healthy, 2 = Normal, 3 = Slightly Unhealthy, 4 = Unhealthy, 5 = Very Unhealthy). As a demonstration of the classification conditions mentioned in Section 3.7, the classification process considered both usage time and neck angle. Previously, Fig. 13

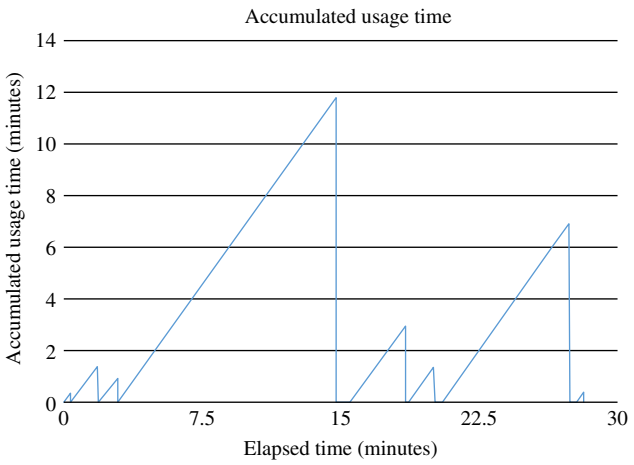


Fig. 13. Usage time in minutes

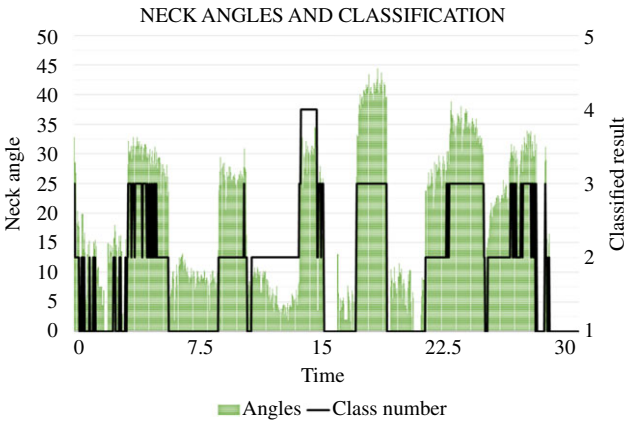


Fig. 14. Neck angles and classification result using pre-defined conditions

showed that the tracked usage time at 15th minute was between 10 and 12 min. The neck angle calculated at the same point of time was between 30° and 35°. If the calculation were to be based on only neck angle, the result would be classified as 'Slightly Unhealthy', but Fig. 14 shows that the classified result was labeled as 'Unhealthy' (4). This was because the usage time was between 7.5 and 15 min and the classification result will worsen by one step based on the classification conditions in Table III. At other points of time where usage time was not more than 7.5 min, the classification was only based on the neck angle.

5. Conclusions

In this paper, we discussed and demonstrated a new method of measuring the neck angle to prevent syndromes that could occur from unhealthy neck postures while using a smartphone. The method uses a combination of data from smartphone sensors and data from an image detection method to calculate the neck angle. The system calculates the neck angle based on those obtained values, which enables the system to give high-accuracy measurements even if the face plane is not parallel to the smartphone screen. Further, we set up an experiment to test the algorithm with a total of 30 human subjects, consisting of multiple nationalities and ages. The results from the paired sample t-test showed that there was no significant difference in the neck angle between the smartphone monitoring system and the photogrammetric method. This supports our assumption and confirms that our proposed algorithm for the neck angle calculation is feasible. Finally, we presented a neck posture classifier that classifies user's neck posture based on both the neck angle and the prolonged usage time. This classification model can help users avoid getting text neck syndrome, which is caused by both the position of the neck and prolonged usage of a smartphone.

However, there are system limitations that need to be addressed. Despite the proposed algorithm and the positive result of the experiment, we cannot overlook that, in theory, the algorithm requires the user's line of sight to be perfectly perpendicular to the smartphone camera. Even though we have shown that the result from the experiment might be positive and that the experiment was done without fixing the subjects' line of sight, we still consider this issue to be a system limitation. Another limitation is that our algorithm focused on detecting neck angle on pitch direction. Thus, it is not capable of calculating the neck angle when the yaw direction has changed. However, the pitch direction is considered the normal way of how people operate their smartphone.

Other limitations may include the use of the smartphone camera. The smartphone camera's battery consumption should be considered prior to implementation. More advanced battery technologies could solve this issue. In addition, advancements in the world of image processing, image detection, and deep learning will provide the system with even more advanced techniques for detecting facial features, which will overcome the difficulties of detecting images in some extreme conditions.

Overall, we have presented a technique that can accurately measure the user's neck angle. Because our system uses resource from the smartphone itself, it is more flexible than other related methods. The system classification provides an easy-to-understand result for the user. The statistical usage record helps a user understand his or her behavior over a certain period. Therefore, it is easier for users to improve their behavior. The image detection technique makes it possible to calculate the face angle, which enables calculations of the neck angle with higher accuracy when the user's face plane is not parallel to the smartphone plane. Most importantly, the experiment showed a positive result when comparing the results of the proposed method with the results from the previously published photogrammetric method. Despite several limitations, we have shown that the system is feasible and it can help people improve their behavior when using a smartphone and prevent syndromes like text neck.

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